



ARTIFICIAL INTELLIGENCE





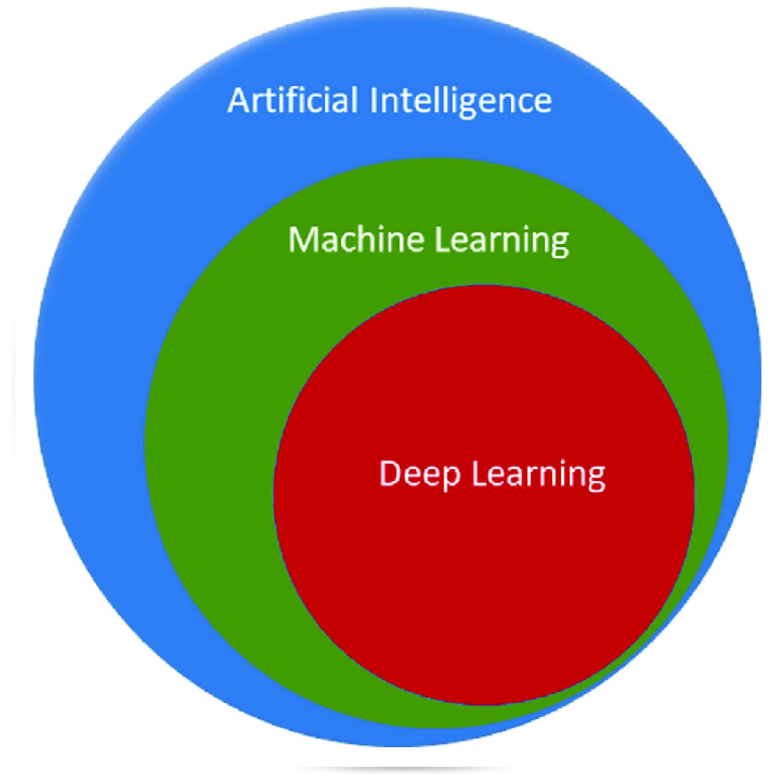
Advanced Concepts of Modeling in Artificial Intelligence

2.1.Revising AI, ML and DL

Differentiate between AI, ML and DL

- To develop an AI-based project, **it's essential to work with Artificial Intelligence models or algorithms.**
- This can be achieved either by **creating a custom model** or **by utilizing existing AI models.**

Let's first understand the definitions of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL).



1.Artificial Intelligence

Artificial Intelligence, *refers to any technique that enables computers to mimic human intelligence*. An artificially intelligent machine works on algorithms and data fed to it and gives the desired output.

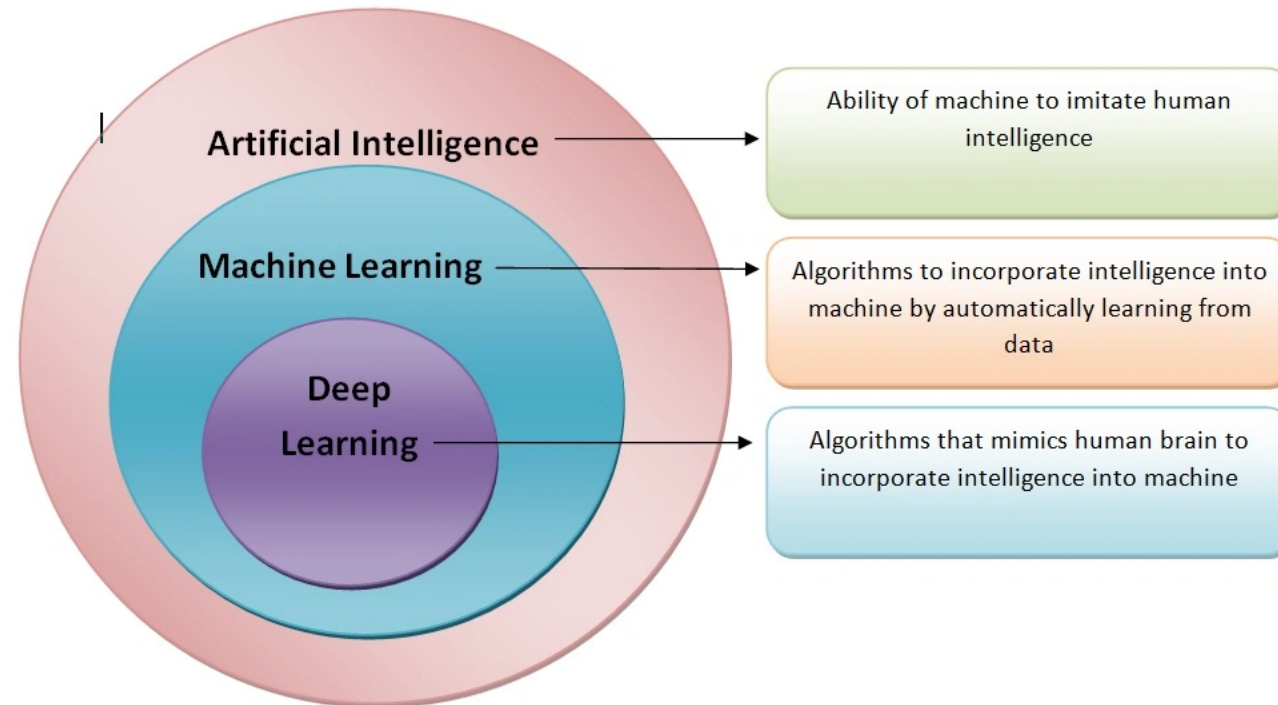
2.Machine Learning

Machine Learning, *enables machines to improve at tasks with experience*. The machine here *learns from the new data fed to it* while testing and uses it for the next iteration

3.Deep Learning

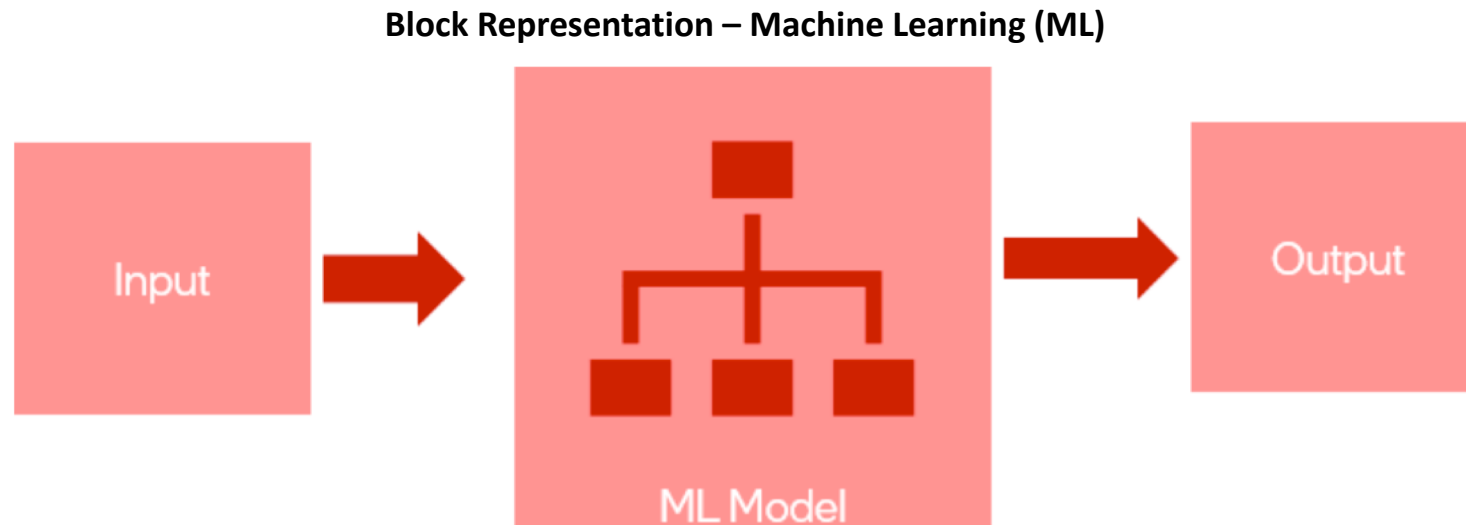
Deep Learning, *enables software to train itself to perform tasks with vast amounts of data*. Since the system has got huge set of data, it is able to train itself with the help of multiple machine learning algorithms working altogether to perform a specific task.

Refer below Venn Diagram, Artificial Intelligence is the ***umbrella terminology which covers machine and deep learning*** under it and Deep learning comes under Machine learning.



Machine Learning - ML

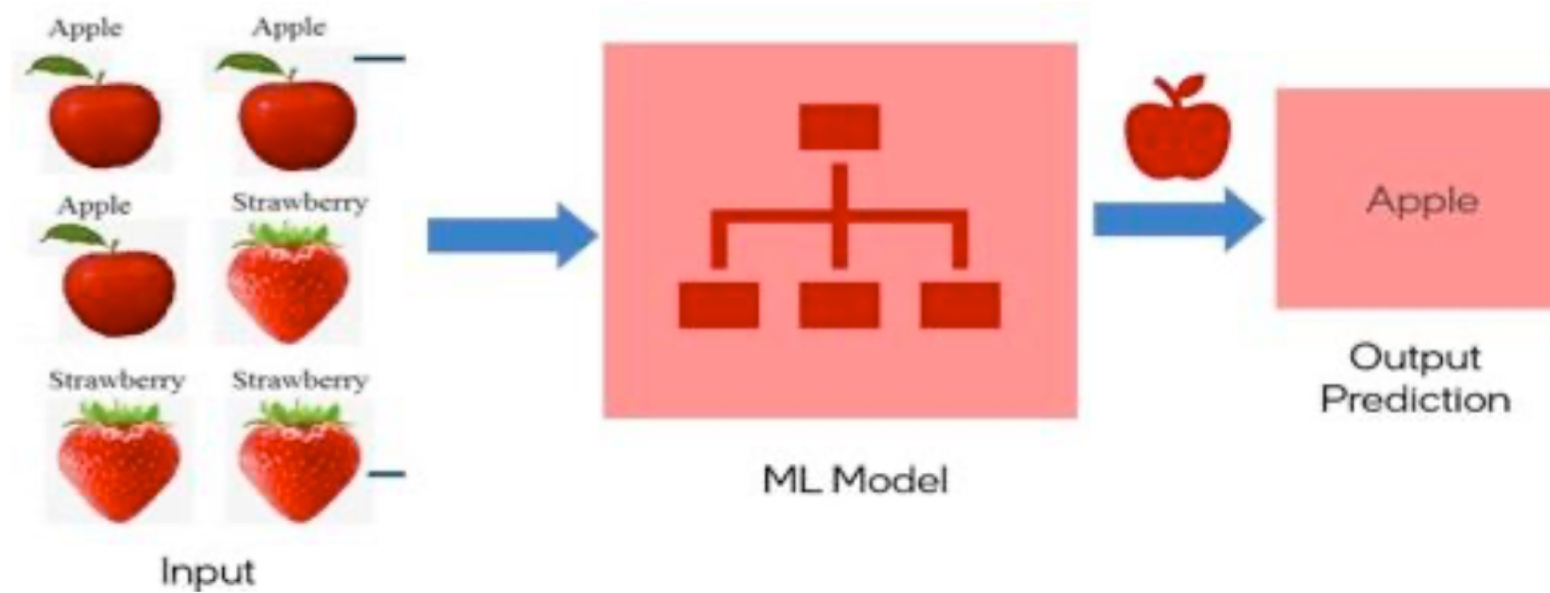
Machine Learning, or ***ML enables machines to improve at tasks with experience.*** The machine ***learning from its mistakes and takes them into consideration in the next execution.*** It improves itself using its own experiences.



Input (past or historical data) is given to the ML model and the model generates output by learning from the input data.

Example

Shows labelled images (every image is tagged either as apple or strawberry) are given as input to the ML model. Model learns from the input data to classify between apples and strawberries and predicts the correct output as shown.



Object Classification

Example

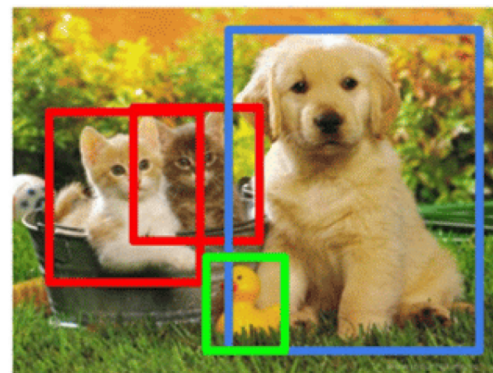
Identifies and labels objects present within an image or data point. It determines the category an object belongs to.

Classification



CAT

Object Detection

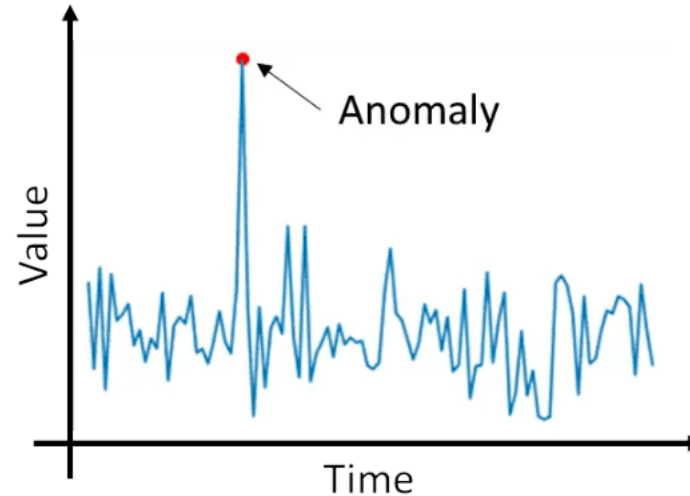


CAT, DOG, DUCK

Anomaly Detection

Example

Anomaly detection helps us find the unexpected things hiding in our data. For example, tracking your heart rate, and finding a sudden spike could be an anomaly, flagging a potential issues.

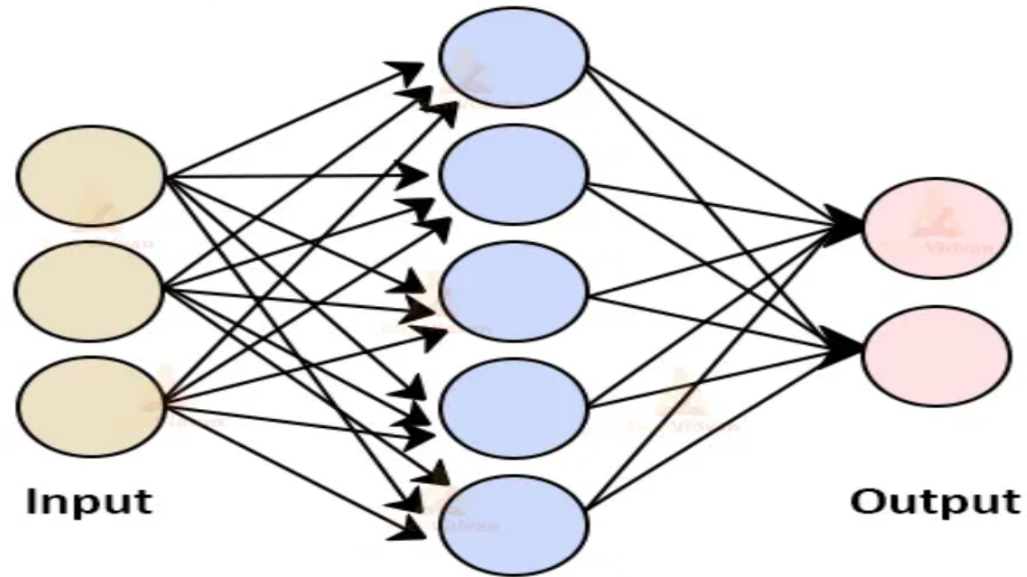


Deep Learning - DL

DL enables to ***train itself to perform tasks with vast amounts of data***. In Deep Learning, the machine is ***trained with huge amounts of data which helps it into training itself around the data***.

- Such Machines are intelligent enough to develop algorithms for themselves.
- Deep Learning is the most advanced form of AI.

Block Representation- Deep Learning(DL)

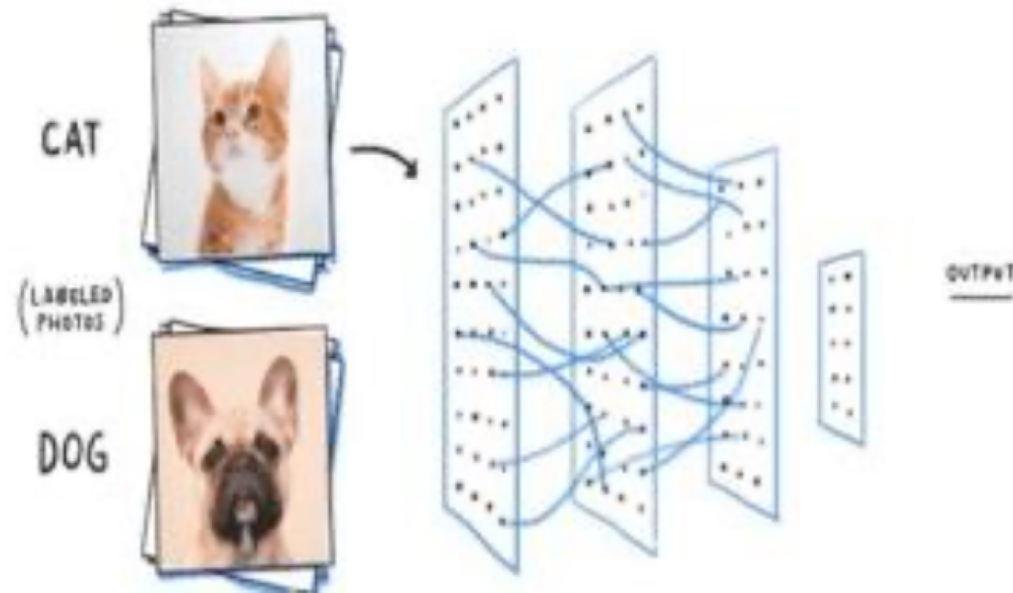


Input is given to an ANN, and after processing, the output is generated by the DL.

Object Identification

Example

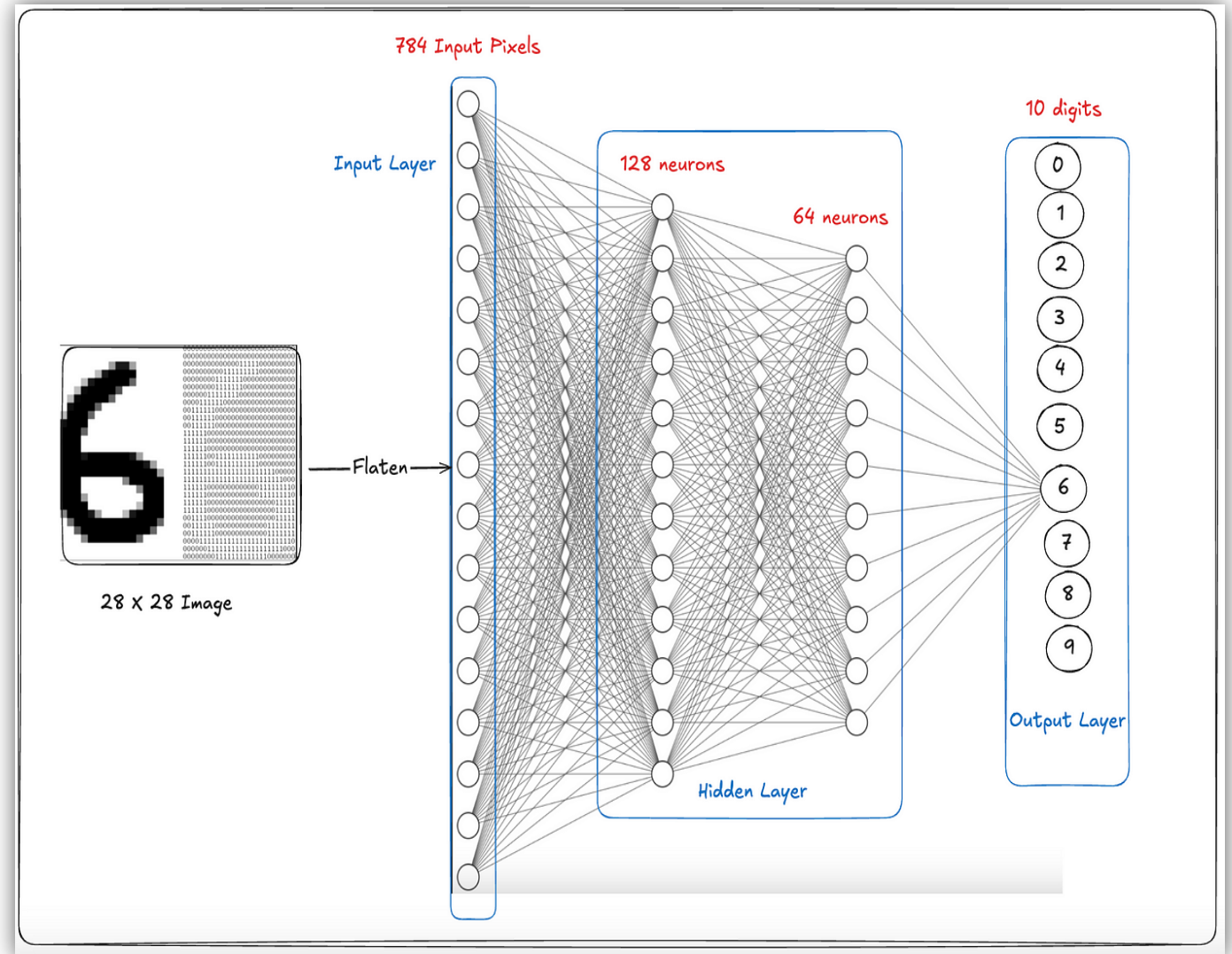
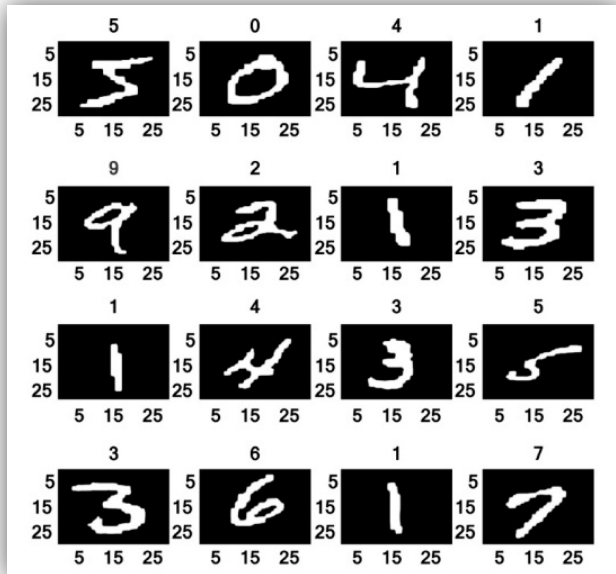
Object classification in deep learning tackles the tasks of identifying and labeling objects within an image. It essentially uses powerful algorithms to figure out what's in a picture and categorize those things.



Digit Recognition

Example

Digit recognition in deep learning *tackles the challenge of the training computer to identify handwritten digits (0-9) within images*



2.1 Common terminologies used with data

What is Data?

- Data is **information** in any form
- Example, A table with information about fruits is data
- Each row will contain information about different fruits
- Each fruit is described by certain features.

Fruit	Color	Price
Apple	Red	\$1.8
Orange	Orange	\$2
Banana	Yellow	\$1
Grape	Purple	\$3

What do you mean by Features?

- Columns of the tables are called **features**
- In the fruit data example, features may be *name, color, size* etc.,
- Some features are special, they are called **labels**



Fruit	Color	Price
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2.1 Common terminologies used with data

What are Labels?

Data Labeling is the process of attaching meaning to data

- It depends on the context of the problem we are trying to solve.
- For example if we are trying to predict **what fruit it is based on the color**, then **color is feature**, and **fruit name is the label**.



Fruit	Color	Price
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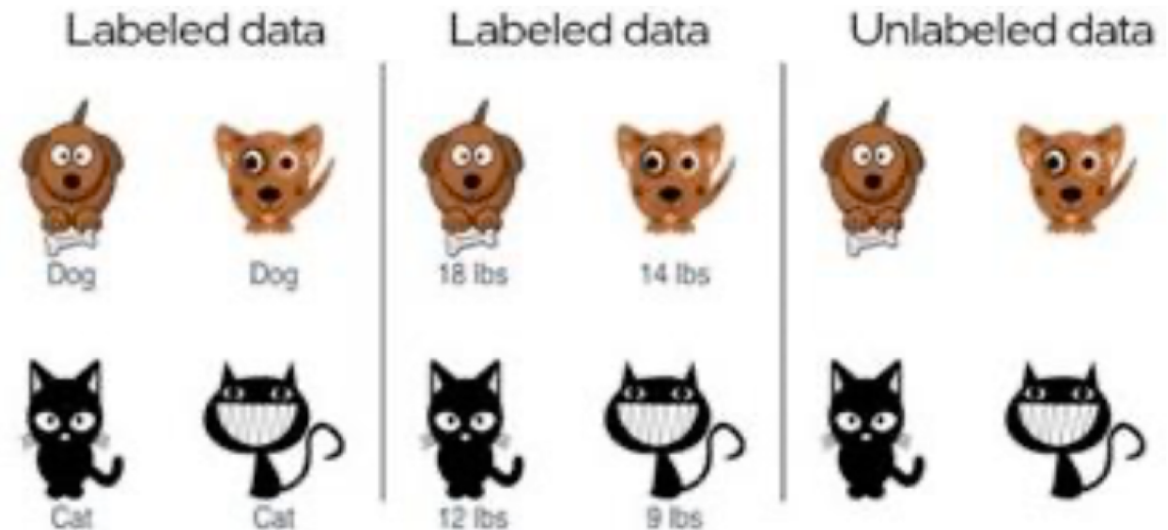
2.1 Common terminologies used with data

Labeled Data

- Data to which some **tag/label** is attached.
- For example Name, type, number etc.

Un-Labeled Data

- The **raw** form of data
- Data to which no tag is attached



2.1 Common terminologies used with data

What do you mean by a training data set?

- The training data set is a ***collection of examples given to the model to analyze and learn.***
- Just like how a teacher teaches a topic to the class through a lot of examples and illustration.
- Similarly, a set of labeled data is used to train the AI model.

What do you mean by a testing data set?

- The testing data set is ***used to test the accuracy of the model.***
- Just like how a teacher takes a class test related to topic to evaluate the understanding level of students.
- Test is performed without labeled data and then verify results with labels.

